WaveNET Stage D Report: Training the U-Net Baseline

Summary

Stage D trained a U-Net baseline that scores test IoU 0.783 ± 0.014 (3 seeds) on M09, a tectonic family never seen in training. Velocity error is about 0.02 km/s. No tuning we tried (learning rate, loss weight, decoder) moved validation IoU or velocity error beyond seed noise.

- **Result:** test IoU 0.783 ± 0.014, Dice 0.878 ± 0.009, velocity RMSE 0.0167 ± 0.0015 km/s (mean of per-batch values; pooled over all rows it is about 0.025). Test was scored once per seed.
- **Hardest case:** validation family M02 (IoU 0.560); the model follows the theory curve to about 12 s, then drifts low.
- **Main limit:** train IoU is about 0.87 against 0.64 on validation. The model fits its 7 training families better than it generalizes. The test claim rests on a single held-out family.
- **Conclusion:** the remaining gap is a cross-family generalization problem (data diversity), not an optimization or decoding problem.

Setup

Stage D trains a binary segmentation model that marks the dispersion-curve ridge in an FTAN image, using the cache and batching built in Stages A to C.

Item	Value
Data	20,000 synthetic samples (full production cache, 3.75 GB HDF5)
Input / target	FTAN image and binary curve mask, each 80 × 300 (76 real period rows, 4 padded)
Positive pixels	about 1.67% of each mask
Split (by family)	train M01, M03–M08 (14,000); val M02, M10 (4,000); test M09 (2,000)

Item	Value
Model	UNetSeg, features (16, 32, 64, 128), 1,943,761 parameters
Loss	CombinedLoss: Focal + Dice + Weighted BCE (pos_weight 15) + Sharpening
Optimizer	AdamW, ReduceLROnPlateau (halving), early stopping, batch size 16
Hardware	Bluehive3 SLURM, coe partition, 1 H100 per task

The split is by tectonic family, so no family appears in more than one split. This is what makes the test score a statement about unseen Earth structure, and also why it rests on one family.

Pipeline work before training

Stage C was verified on the full cache (875 / 250 / 125 batches, no split leakage, all range checks passed), but two problems blocked real training.

- **HDF5 reopen bottleneck.** The dataset reopened the HDF5 file on every sample. It now reads metadata once and keeps one h5py handle per worker process, opened lazily and keyed by PID; the handle is dropped when the dataset is pickled so each worker opens its own.
- **Worker recreation.** DataLoader workers were rebuilt every epoch. `persistent_workers=True` is now the default. Together with the handle fix, epoch time fell from 35.9 s to about 3 s in the local test.
- **U-Net width.** The width goes 300 → 150 → 75 → 37 → 18, which does not halve cleanly, so the decoder keeps a defensive interpolate to the encoder resolution.
- **Cluster setup.** Group access, the venv, and the data move to Bluehive3 scratch were completed; jobs run on the coe partition only (a combined `-p coe,h100` request is rejected by Slurm).

Training code and learning-rate sweeps

The sweeps did not separate the learning rates beyond seed noise, so 3e-3 was chosen for stability, not because it was clearly best.

- **`train.py`**. AdamW, plateau LR halving, early stopping, best and last checkpoints, resume. `--lr` has no default, so every run states its rate. Each epoch logs train and val loss, IoU, Dice, velocity RMSE, and a per-family breakdown to `train_log.jsonl` and `train_log.txt`.
- **`eval_by_family.py`**. Re-scores any checkpoint per family on val.
- **Sweep 1** (job 2019070, short runs) and **sweep 2** (job 2091906, 100 epochs, plateau patience 5, early-stop patience 15) compared learning rates on the same split. The sbatch script uses 1 H100 and 3 workers per task, a 1-hour limit, no requeue, and refuses to overwrite an existing output directory.
- **Outcome.** The IoU peaks of 1e-4 and 3e-3 tie. 3e-3 trains more stably and reaches a velocity error of roughly half that of 1e-4 over the same stretch (about 0.05 km/s for 1e-4). Seed spread (±0.02 val IoU) is about as large as the gaps between learning rates, so the sweep supports "3e-3 trains stably", not "3e-3 is best".
- **Earlier misreading, corrected.** Stage B's concern that 1e-4 was too slow came from a 1-sample overfit test; at full scale 1e-4 trains fine. The job IDs 2023906/7 in the sweep-1 configs are Slurm array-task IDs, not reruns; every run started clean.

Final run: 3 seeds at LR 3e-3

Job 2107428 trained seeds 42, 43 and 44 for about 22 minutes each. Each seed's selected checkpoint was scored on val and, once, on test (M09).

Metric	Seed 42	Seed 43	Seed 44	Mean ± sd
Best epoch	67	92	77	
Val IoU	0.625	0.665	0.632	0.641 ± 0.021
Val IoU, M02	0.547	0.590	0.543	0.560 ± 0.026
Val IoU, M10	0.712	0.745	0.731	0.729 ± 0.017
Val vRMSE (km/s)	0.0226	0.0202	0.0232	0.0220 ± 0.0016
Test IoU (M09)	0.767	0.792	0.789	0.783 ± 0.014
Test Dice (M09)	0.868	0.884	0.882	0.878 ± 0.009
Test vRMSE (km/s)	0.0172	0.0151	0.0179	0.0167 ± 0.0015

Test scores higher than val, most likely because M09 is an easier family (no leakage was found), and val is half M02, the hardest. Predicted masks are 6–13% wider than the truth on every seed and family. Seed 43 was still improving slightly at the 100-epoch cap; the others stopped earlier. Test must not be re-scored after further tuning.

Metric definitions and diagnostics

The velocity error is measured against the theory curve, not the FTAN energy ridge, and the headline value depends on how it is averaged.

- **Predicted curve.** For each period row, the probability-weighted mean velocity over all 300 bins of the raw model output. It is not taken from the thresholded mask, and faint background probability pulls it.
- **Reference.** The theoretical fundamental-mode group-velocity curve, entering through the target mask (a 5-bin band around the curve), so it is the theory curve rounded to the 0.01 km/s bin grid.
- **Periods.** All 76 rows, 1.00–19.75 s in 0.25 s steps, unweighted. Val and test have no rows missing a theory value; only 2 of 14,000 training samples do.
- **Averaging.** The logged value is the RMSE per batch of 16, averaged over batches (val 0.022, test 0.017). Pooled over all rows it is about 0.024 on val and 0.025 on test, so the error is concentrated in a minority of samples, especially on test. Quote it as about 0.02 km/s with the definition.
- **Held-out analysis** (job 2107750, `heldout_analysis.py`). Scores every val and test sample per seed and saves example predictions; it reproduced every logged value exactly.
- **Mask width.** The 5–7% over-width also appears on training data (train IoU about 0.87), so it is not a generalization effect.

Poster figures (white background, 300 DPI) are in `docs/ml_pipeline_stages/figures/`: training curves for the three seeds, per-family IoU bars, and three example predictions chosen by IoU percentile (easy: M10, 90th percentile, IoU 0.88; typical: M09, 50th, 0.83; hard: M02, 10th, 0.37).

Ablation: pos_weight and decoder

No configuration beat the baseline on val IoU or val velocity error. Job 2109273 (code commit e676a7c) ran 9 tasks on val only: A re-scored the existing pos_weight 15 checkpoints, B retrained at pos_weight 5, C at pos_weight 1, each with seeds 42/43/44.

Test was not scored.

Config	Val IoU	pred / true width	vRMSE soft, per-batch / pooled (km/s)	vRMSE peak, per-batch / pooled (km/s)
A: pos_weight 15	0.641 ± 0.021	1.108 ± 0.023	0.0220 / 0.0244	0.0224 / 0.0249
B: pos_weight 5	0.643 ± 0.012	1.047 ± 0.013	0.0223 / 0.0245	0.0227 / 0.0250
C: pos_weight 1	0.614 ± 0.024	0.983 ± 0.002	0.0296 / 0.0344	0.0314 / 0.0383

- **Per-row-peak decoder** (argmax bin per row, then a weighted mean over ±3 bins): equal to or slightly worse than the soft centroid in every configuration, within seed noise. The error comes from where the band sits, not from how spread out the probability is.
- **pos_weight 5** roughly halves the width bias (1.11 to 1.05; per-seed ranges do not overlap) with no change in IoU, Dice or velocity error.
- **pos_weight 1** removes the bias but is a trade-off: velocity error rises from 0.022 to 0.030 (every C seed worse than every A seed) and M10 IoU falls from 0.729 to 0.687.
- **Convergence.** B and C ran to the 100-epoch cap and were flat at the end (at most 0.007 IoU gained over the last 20 epochs).

Findings, open questions and next steps

The baseline is established and robust to hyperparameters; the open problem is generalization across families.

What the results support

- The pipeline works end to end, from raw simulation to a trained model that picks dispersion curves on an unseen family.
- Seed-to-seed spread is small (test IoU ±0.014), so the result is reproducible.

What they do not support

- "Generalizes to unseen tectonic settings" rests on one test family, and val (2 families) was also used to select checkpoints.
- The LR sweep did not show 3e-3 is best, only that it is stable.

- Single-run comparisons of small tweaks cannot be trusted; seed noise (about ±0.02 val IoU) is as large as the effects tested.

Questions

1. Test is a single family (M09), and it looks easier than val. Should the split design change, or should cross-validation over family combinations be added?
2. The train/val gap (IoU 0.87 vs 0.64): is more earth-model diversity (more families or wider parameter ranges) acceptable? This touches the simulator and needs her sign-off.
3. Mask width bias of 6–13%: adopt pos_weight 5 as the default? That means retraining and re-scoring test once, as a deliberate and disclosed step.
4. batch_size 16 was never formally decided.
5. Should the tutorial sample get split support?

Next steps

- Optional experiment that needs no simulator change: augmentation (amplitude scaling, noise, random masking) aimed at the train/val gap.
- Check whether FTAN energy at long periods is weak for M02; if so, the drift past 12 s is a data limit to report, not a model failure.
- Stage E: prediction and evaluation.